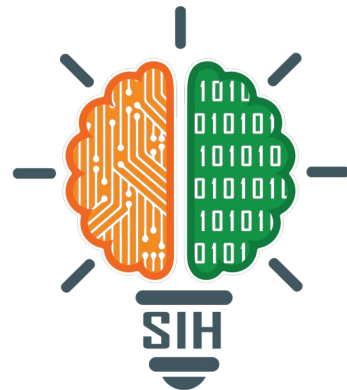


# SMART INDIA HACKATHON 2026



## TITLE PAGE

- Problem Statement ID – SIH26097
- Problem Statement Title – AI-driven voice assistant for livelihood mapping and NSQF-aligned skilling recommendations for SC communities under the GIA component of PM-AJAY
- Theme – Agriculture, FoodTech & Rural Development
- PS Category – Software
- Team ID – \_\_\_\_\_ (fill from the SIH portal)
- Team Name (Registered on portal) – Cognify



# SMART INDIA HACKATHON 2026

**Wrong course, then no job**

that is what happens to SC families today (SIH26097 · MoSJE)

PM-AJAY pays for skill training and small-business support (up to ₹50,000). But people are picked and placed through paper forms and field visits — those who can't read, or don't know the trades, land in the wrong course.

**41% got a job**

after government training

Source: CAG audit 2025

**43% have no phone**

of rural women, of their own

Source: NFHS-6, 2023-24

**34% cannot read**

of SC adults

Source: Census 2011

**WHO WE SERVE** 20.14 crore SC people → ≈ 5.9 crore below the poverty line (eligible) → ≈ 50,000 get GIA support each year → < 1 in 1,000 a year → every seat must fit

Census 2011 (via MoSJE Handbook 2021) · SC poverty ratios 31.5% rural / 21.7% urban (Planning Commission, 2011-12) · GIA: 2,01,006 beneficiaries in 2021-25 → ≈ 50,000 a year (PIB, 19 Mar 2025)

**ONE CALL, SIX STEPS** about 10 minutes, on any keypad phone

1

**Missed call**

he gets a free call-back

Within a minute, from a govt 1600 number, in the language he picks on the keypad.



2

**Facts by keypad**

so a speech error can't hurt

Age, education, travel distance are pressed on keys — eligibility never depends on speech.



3

**He talks, AI listens**

his job becomes an NCO code

"Aap kya kaam karte ho?" The AI finds his official occupation and asks about his skills.



4

**Read back & saved**

one clean profile, no paper

He hears everything back and presses 1 to confirm, 2 to correct.



5

**1-4 options that fit**

every fact checked first

Only courses, schemes or jobs near him that truly fit. He can ask any doubt.



6

**Apply & follow-up**

CSC → officer → 3 calls

SMS with Case ID + nearest CSC; officer approves in the AJAY app; calls at 1, 3 and 6 months.

**WHAT MAKES IT DIFFERENT** six things a paper form can never do**Smart questions**

each depends on the last answer  
Bike mechanic → engine, tyres, oil  
— not a generic form.

**Profile writes itself**

officers only review and approve  
The AI fills the full profile table in the proper format.

**Only verified govt data**

nothing invented, ever  
Rules, NQR courses, NCO codes, centres — loaded by us, never the internet.

**SATYA fact-check**

no match in the DB → not said  
Every fact is matched to a database row before it is spoken.

**Cheapest AI first**

bigger model, then a human, only if needed  
Common answers get cached — cost falls over time.

**Web portal too**

same case, no call needed  
Mobile + OTP login for the family, CSC or officer to view or edit.

**Also answers the 5 GIA issues in the brief:** planning · beneficiary selection · placement after training · coordination · too few ground staff

## TECH STACK Indian, open where possible

### Calls & voice

Exotel / Plivo

WhatsApp API

Pipecat voice server

### AI models

Saaras v3 • STT

Bulbul v3 • TTS

Sarvam-30B / 105B

IndicConformer

Bhashini

SATYA (ours)

### Backend & data

Python

FastAPI

Celery

Redis

PostgreSQL

pgvector

### Apps & govt links

React

Leaflet maps

NCO / NQR

AJAY app\*

DigiLocker\*

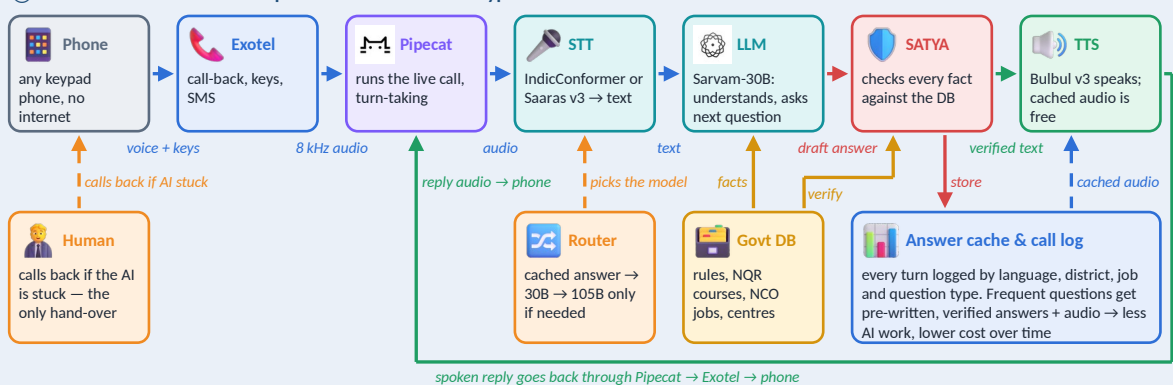
Docker • MeghRaj\*

\* needs a government MoU.

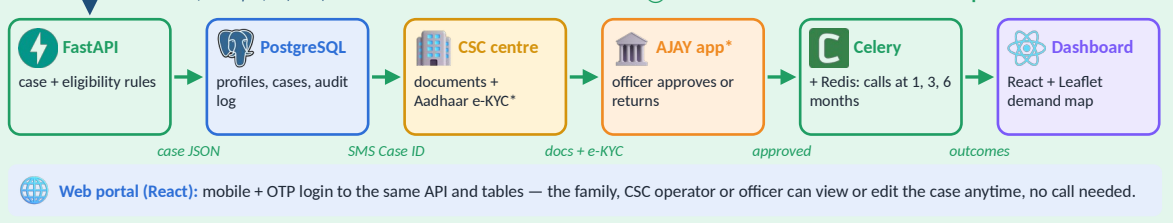
[Full architecture, API docs and code](#) → [GitHub repo](#) (add link)

## HOW IT WORKS two lanes: during the call (AI, in seconds) and after it (people, in days)

### ① DURING THE CALL — each spoken turn = 1.2-2.2 s • keypad turns are instant



### ② AFTER THE CALL — the confirmed profile becomes a case



AI understands → verified data decides → SATYA checks → humans approve

Data rules: no Aadhaar number stored · raw audio deleted after confirmation · consent logged (DPDP Rules 2025) · every edit in the audit log

**Can it be built and run?** Yes — every part already runs in India, the prototype can be built today, and one completed call costs about ₹25–54.

**ALREADY WORKING IN INDIA** *nothing here is untested*



**10 languages out of 22**

are already in use

UIDAI already runs Sarvam's voice AI for Aadhaar on its own servers, in 10 of Sarvam's 22 languages.

Source: PIB / MeitY 2025



**12 lakh calls a day**

already handled by a govt line

Kilkari, the government health-advice line, runs this volume every single day.

Source: BMJ Global Health 2021



**≤ 20% word error**

on real Indian phone calls

Best Indian speech AI, tested on 36,691 speakers in 15 languages (536 hours).

Source: Voice of India 2026 (arXiv)



**1.2–2.2 s per reply**

fast enough to feel like a talk

End of speech 0.5 s + speech-to-text 0.1–0.3 s + AI first word 0.3–1 s + voice 0.2 s + network 0.1–0.2 s.

Source: Sarvam docs • NVIDIA 2026

**ONE COMPLETED CALL COSTS ₹25–54** *per person, all-in: phone + AI + SMS + human help*  
Completed call = 10-min interview + options, plus 3 follow-up calls of 2 min (= 16 min of phone time).

| Cost item                          | How we calculate it                                    | Low          | High         |
|------------------------------------|--------------------------------------------------------|--------------|--------------|
| Phone line (Exotel)                | 16 min × ₹0.80–1.80, + unfinished calls & 30-s billing | ₹14.7        | ₹33.7        |
| Speech → text (Saaras v3)          | ₹30 per hour; only his speech (50–60% of the call)     | ₹4.2         | ₹5.1         |
| Text → speech (Bulbul v3)          | ₹3 per 1,000 characters; fixed questions pre-recorded  | ₹2.4         | ₹6.0         |
| AI brain (Sarvam-30B / 105B)       | ~90% of turns on 30B; 105B only when needed            | ₹0.25        | ₹0.5         |
| SMS + human call-back              | 4–6 SMS; 10–20% of callers need a human for 5–6 min    | ₹2.2         | ₹5.3         |
| Hosting & helpline number          | servers, database, number — spread per call            | ₹1.0         | ₹3.0         |
| <b>TOTAL per completed profile</b> | <b>1,000 missed calls → 420–517 finished profiles</b>  | <b>≈ ₹25</b> | <b>≈ ₹54</b> |

**Cost falls further over time** cached answers, cheaper cached tokens, free self-hosted speech  
Frequent questions get pre-written, verified answers and audio (answer cache, slide 3); Sarvam bills cached input at ₹10.98 vs ₹29.28 per 1M tokens; at national scale ~70% of speech moves to free IndicConformer → ₹21–49 per profile.

Prices: Sarvam API pricing • Indian telephony rate cards 2026 • funnel = our estimate

#### 4 RISKS, EACH WITH AN ANSWER

**Village dialects are hard** → keypad for facts • read-back • bigger model • then a human

**AI could say something wrong** → SATYA blocks anything not in the govt database

**People ignore unknown numbers** → he calls first; call-back from an official 1600 number

**Slow replies bore callers** → streaming voice • pre-recorded questions • keypad turns are instant

**76% OF CALLS ANSWERED BY THE 3RD TRY** *and only 10–20% ever need a human*

- 50% answer on the 1st attempt, 76% by the 3rd, 99.5% by the 9th
- ~80% of answered calls finish the interview; ~85% of those finish the options → 420–517 profiles per 1,000 missed calls
- 10–20% need a human counsellor call-back — design target ≤ 20%
- Can't be reached by phone? The web portal (mobile + OTP) is always open.

Source: Kilkari, BMJ Global Health 2021

Source: our funnel estimate

Source: our design target

**THE GOVERNMENT CAN AFFORD IT** *less than ₹1 of every ₹100 it already spends*



**0.05–0.10% of the budget** is all SkillCall needs

All of India: ₹1.1–2.1 crore a year vs PM-AJAY's ₹2,140 crore (2026–27).



**Under ₹1 in every ₹100** of what each beneficiary already gets

≈ ₹220–420 per beneficiary vs ≈ ₹60,700 average GIA support (₹1,219.80 cr ÷ 2,01,006).



**₹9–13 lakh to start** a 6-month pilot in one block

≈ 1,000 families: build, test with real users, then scale to the state.

**Who gains?** SC families, CSC operators, district officers and the Ministry. Every number shows how it was worked out.

**WHAT CHANGES** *four numbers, each with its working*



**41% → 70% placed**

the gap we close

Only 41% of trained people get a job today; 70% is the govt's own target. Per 1,000 trainees, even 50% means 90 more people working.

Source: CAG 2025 · MSDE norms



**62% → 73% stay in the job**

when told the truth before joining

+11 percentage points (men) in a field trial of two honest job-information sessions. SkillCall gives that on the call.

Source: J-PAL · Chakravorty et al. 2024



**30 h → 5 h a month**

of officer time, per 40 cases

Today 45 min each (interview + 3 follow-up calls); with SkillCall a 7.5-min review. India:  $50,000 \times 37.5 \text{ min} = 31,250 \text{ h a year} \approx 18 \text{ full-time officers}$ .

Our estimate: AJAY app's data entry not counted



**₹12.5–15 cr a year**

not wasted on wrong matches

If wrong matches fall in 5 of every 100 cases:  $2,500 \times ₹50,000 = 60,700 \text{ support} = 6-14 \times \text{SkillCall's yearly cost}$ .

Our estimate: PIB 2025 average support

**1–1.5 lakh profiles a year** so every one of the  $\approx 50,000$  GIA seats goes to the person it fits — 2–3 applicants screened per seat (calculations sheet \$1.9)

**WHO WINS** *and what each one actually gets*



**SC family**

right course, first time

Free, in his own language — and calls until he is earning.



**CSC operator**

the case arrives pre-filled

Only documents and Aadhaar e-KYC are left to do.



**District officer**

ready, checked cases

In the AJAY app, plus a demand map by block for the yearly plan.



**Ministry**

one record, call to job

Each outcome marked confirmed (employer / EPFO\*) or self-reported.

**BENEFITS** *social · economic · governance · environment*



**Social**

Women and non-readers included; right course the first time.



**Economic**

Less money on wrong training; more people earning — proved by follow-up.



**Governance**

One shared case for CSC, officer, partner and bank; planning data, not paper.



**Environment**

No paper forms, fewer trips to offices, runs on the shared govt cloud.



## Government data

PIB, Rajya Sabha reply (19 Mar 2025) — GIA: 2,01,006 beneficiaries, ₹1,219.80 cr [pib.gov.in](https://pib.gov.in)

PIB PM-AJAY explainer (May 2025) — eligibility, 5% admin & MIS line [pib.gov.in](https://pib.gov.in)

CAG Report No. 20 of 2025 — PMKVY: 41% placed, no skill-gap analysis [cag.gov.in](https://cag.gov.in)

NFHS-6 (2023-24) — women's phone ownership 63.6%, rural 57.4% [rchiips.org](https://rchiips.org)

Census 2011 (MoSJE Handbook 2021) · SC poverty ratios 2011-12 (Lok Sabha) [socialjustice.gov.in](https://socialjustice.gov.in)



## Technology

Sarvam API pricing & docs — Saaras v3, Bulbul v3, 30B / 105B; voice-agent guide [docs.sarvam.ai](https://docs.sarvam.ai)

NVIDIA × Sarvam 2026 — Sarvam-30B serving, first token < 1 s (P95) [developer.nvidia.com](https://developer.nvidia.com)

Artificial Analysis — Sarvam-30B ≈ \$0.03 per 1M tokens (blended) [artificialanalysis.ai](https://artificialanalysis.ai)

AI4Bharat IndicConformer · Pipecat · Bhashini [ai4bharat.iitm.ac.in](https://ai4bharat.iitm.ac.in)



## Research

Chakravorty et al. 2024 / J-PAL — honest job info: 62% → 73% stay in job [povertyactionlab.org](https://povertyactionlab.org)

Bashingwa et al. 2021, BMJ Global Health — Kilkari call answer rates [gh.bmj.com](https://gh.bmj.com)

Voice of India 2026 — speech-AI benchmark on real Indian phone calls [arxiv.org](https://arxiv.org)

Medhi et al. 2011, ACM TOCHI — voice interfaces for low-literacy users [dl.acm.org](https://dl.acm.org)



## Standards & policy

PM-AJAY portal & AJAY app launch (26 May 2026) — PIB [pib.gov.in](https://pib.gov.in)

NCO-2015 occupation codes · NQR / NCVET job roles [nqr.gov.in](https://nqr.gov.in)

DPDP Rules 2025 — consent and data protection [meity.gov.in](https://meity.gov.in)

TRAI 1600-series rules for government calls (2026) [traf.gov.in](https://traf.gov.in)

MeitY (Lok Sabha 2023) — 5.58 lakh CSCs, 4.44 lakh at gram panchayats [eparlib.nic.in](https://eparlib.nic.in)

**SEE MORE** *code, prototype, video and every calculation — one click each*



### GitHub repo

code, architecture docs, API reference

[\(add link\) →](#)



### Live prototype

try a SkillCall conversation in the browser

[\(add link\) →](#)



### Demo video

2-minute walkthrough of a real call

[\(add link\) →](#)



### Calculations sheet

every number, its source and full working

[\(add link\) →](#)